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SARE-CD: Structure–Appearance Routed Experts for Building Change Detection in Remote Sensing Imagery

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Highlights

What are the main findings?

- A front-end statistical correction combined with pixel-wise routing over Structure, Appearance, and Adaptive Experts lets geometric changes and appearance drift receive separate transformations instead of sharing a single discrimination path.
- SARE-CD attains F1 scores of 91.58%, 65.78%, and 84.46% on LEVIR-CD+, S2Looking, and SYSU-CD, exceeding eight representative baselines while using 15.42 M parameters and 20.41 G FLOPs.

What are the implications of the main findings?

- Separating global radiometric drift from local structural difference helps preserve weak boundary cues for small building changes, particularly on S2Looking; however, extreme additive brightness offsets remain challenging and can substantially reduce F1 retention.
- Compact expert branches and per-pixel routing keep the accuracy gain from resting on a large computational budget, which suits large-area building change monitoring under varying illumination and viewing conditions.

Abstract

In building change detection from high-resolution remote sensing imagery, genuine building changes typically appear as local alterations in roof contours and spatial layouts, whereas differences in solar elevation, atmospheric conditions, and sensor calibration introduce cross-temporal appearance drift in unchanged regions. When the two types of difference are jointly encoded in a single feature space, appearance drift tends to dominate the network response. We therefore propose SARE-CD, a dense, pixel-wise, soft-gated heterogeneous mixture-of-experts (MoE) framework that combines scene-level bias correction with functionally distinct expert modeling. At the feature front end, SARE-CD generates affine parameters from the channel statistics of the bitemporal change representation and applies them in residual form to suppress systematic radiometric drift. The Structure, Appearance, and Adaptive Experts then model geometric boundary changes, local appearance perturbations, and composite differences, respectively, and a pixel-wise router assigns expert weights at each location, with routing-entropy and weak load-balancing regularization used to stabilize the division of labor among the experts. On LEVIR-CD+, S2Looking, and SYSU-CD, SARE-CD achieves F1 scores of 91.58%, 65.78%, and 84.46%, all of which are the highest among the compared methods. In addition, controlled perturbation

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experiments show that F1 retention remains high under hue shifts, soft shadows, and small misregistration, whereas extreme additive brightness offsets lead to a notable F1 decline on S2Looking.

Keywords: bitemporal change detection; appearance drift; Structure–Appearance Routed Experts; pixel-wise routing

1. Introduction

Bitemporal remote sensing change detection identifies genuine surface changes by comparing images of the same area acquired at different times [1–5]. It supports applications such as urban expansion monitoring, disaster damage assessment, and land-use updating. In high-resolution imagery, building changes typically appear as local alterations in roof contours, boundaries, and spatial layouts. Images acquired months or years apart rarely share identical solar elevation, atmospheric conditions, or sensor calibration. Consequently, unchanged regions can differ in brightness, hue, and shadow. Such appearance drift often covers larger areas than genuine building changes and is more visually salient.

Deep change-detection networks have evolved substantially in their feature representations. Fully convolutional Siamese architectures established the basic bitemporal learning paradigm through early fusion, feature differencing, and feature concatenation [6]. Dense skip connections and deep supervision were subsequently introduced to preserve multiscale details [7–9]. Transformers model long-range dependencies and cross-temporal interactions [10–12], whereas state-space models capture global context with linear complexity [13,14]. Mask classification, dual-domain learning, and efficient context aggregation improve change representations from complementary perspectives [15–17]. These methods all seek to learn a stronger unified change representation. However, bitemporal differences in high-resolution urban scenes are heterogeneous. Construction and demolition alter geometric boundaries and spatial layouts, whereas illumination, shadow displacement, and sensor bias cause regional or global appearance drift. When both signals enter the same feature space, appearance drift that is spatially extensive and high in contrast can dominate the network response, and small structural changes with weak boundaries may then be suppressed. As shown in Figure 1, the baseline responds to irrelevant cross-temporal differences, whereas SARE-CD concentrates its predictions more closely on genuine change regions. In Figure 1, the left panel shows a bitemporal image pair from LEVIR-CD+. Yellow boxes and their magnified views mark genuine building changes, whereas red boxes and their magnified views mark cross-temporal differences unrelated to the target changes. Red overlays in the right panel indicate regions predicted as changed, and the baseline produces more false change responses in the irrelevant-difference regions.

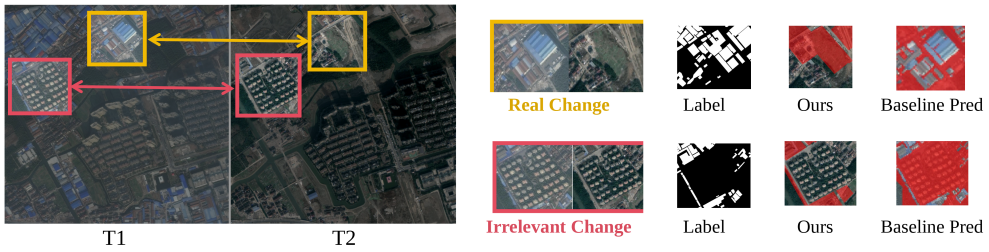

Figure 1. Confusion caused by heterogeneous differences in high-resolution building change detection.

Rather than strengthening the unified change representation, another group of methods acts on the cross-temporal appearance difference itself. Domain adaptation reduces

inter-domain distribution discrepancies and aligns bitemporal features acquired under different conditions [18]. Generative methods treat seasonal and illumination shifts as style factors and learn season-invariant representations through adversarial training [19]. These approaches establish the value of handling appearance shifts explicitly, but they usually require additional generation or disentanglement procedures and therefore increase training cost. Instance normalization and its adaptive variant enable lightweight feature correction, but may suppress local genuine changes [20,21].

To address this problem, we propose SARE-CD, a Structure–Appearance Routed Experts framework. At each scale, SARE-CD constructs a change representation from the two temporal features and their absolute difference. It then derives affine parameters from the representation's channel statistics. A residual correction suppresses systematic shifts caused by global illumination or sensor calibration. The corrected representation enters three specialized experts. The Structure Expert combines a fixed Laplacian edge prior with learnable convolutions to enhance geometric boundaries. The Appearance Expert uses channel recalibration to model brightness, shadow, and texture variations. The Adaptive Expert uses dilated convolutions to capture composite differences not covered by the other priors. A pixel-wise router predicts three weights at each location and fuses the expert responses. To encourage locally confident routing while preventing the overall expert usage from collapsing onto a few branches, we introduce pixel-wise routing-entropy regularization alongside a weak load-balancing constraint.

Our main contributions are as follows.

1. We introduce a front-end statistical correction scheme that derives affine parameters from the change representation's channel statistics.
2. We design a heterogeneous expert pool and pixel-wise routing mechanism. The three experts separately model geometric boundary changes, local appearance perturbations, and composite differences. The router determines each expert's contribution at every spatial location.
3. We introduce routing-entropy regularization to encourage locally confident routing and weak load-balancing regularization to prevent expert collapse, thereby stabilizing the division of labor among experts.
4. On LEVIR-CD+ [22], S2Looking [23], and SYSU-CD [24], SARE-CD achieves F1 scores of 91.58%, 65.78%, and 84.46%, respectively.

In addition, five controlled perturbation experiments show high F1 retention under hue shifts, soft shadows, and small misregistration.

The remainder of this paper is organized as follows. Section 2 reviews remote sensing change detection, pseudo-change suppression, and mixture-of-experts models. Section 3 describes the SARE-CD architecture, bias-corrected change representation, heterogeneous expert pool, and loss functions. Section 4 presents the experimental settings, comparisons, complexity analysis, and ablation studies. Section 5 discusses routing attribution, limitations, and future work. Section 6 concludes the paper.

2. Related Work

2.1. Deep Learning-Based Change Detection Methods

Convolutional networks form the foundation of bitemporal change detection. Early architectures such as FC-EF, FC-SiamConc, and FC-SiamDiff explore early fusion, Siamese concatenation, and Siamese differencing for temporal feature interaction [6]. SNUNet-CD introduces a nested Siamese structure with dense skip connections to fuse features across scales, while Improved UNet++ and DSIFN combine encoder–decoder backbones with deep supervision to refine boundary delineation [7–9]. These designs progressively

strengthen local structural extraction and multiscale reconstruction, yet they operate on raw bitemporal features without using global channel statistics to correct systematic cross-temporal appearance shifts.

Transformers extend the receptive field to model long-range dependencies and cross-temporal interactions [12]. BIT compresses bitemporal features into semantic tokens and captures global spatiotemporal context with a Transformer encoder [10]. ChangeFormer adopts a hierarchical Transformer encoder within a Siamese framework to aggregate multiscale semantics [11]. State-space models achieve comparable global context modeling with linear complexity: ChangeMamba applies a spatiotemporal state-space architecture to binary, semantic, and damage change detection [13], while CDMamba couples global state-space modeling with convolutional detail recovery [14].

While these methods have achieved notable progress, their reliance on global interaction or extended receptive fields broadens contextual modeling without distinguishing the sources of temporal discrepancies. Extensive brightness, hue, or shadow shifts therefore remain entangled with genuine geometric changes in the learned representation. Rather than extending contextual range, SARE-CD redirects attention to the type of temporal difference, allowing each discrepancy to receive a targeted response.

2.2. Pseudo-Change Suppression

Bitemporal remote sensing images are inevitably acquired under differing seasons, solar elevations, sensor states, and atmospheric conditions. As a result, unchanged regions may exhibit systematic discrepancies in brightness, hue, shadow, and texture. Such appearance shifts, often spatially extensive and visually salient, are readily confused with genuine semantic changes and thus constitute a major source of false positives in change detection [1–5,18,19].

Existing suppression strategies can be broadly grouped by the level at which they intervene. At the image level, several studies treat seasonal or radiometric differences as a style factor and employ generative transformation or style-consistency constraints to map bitemporal images into a unified appearance distribution [18–21]. At the feature level, domain-adaptation methods such as DSDANet combine Siamese feature extraction with distribution-matching losses to align bitemporal representations across domains [18]. A more lightweight alternative exploits channel statistics: instance normalization links feature statistics to image style [20], and adaptive instance normalization (AdaIN) aligns per-channel means and variances for style transfer [21], suggesting that low-order statistics are sufficient to characterize and correct global appearance drift.

Despite their respective merits, these methods share a common dilemma. Genuine changes are themselves local bitemporal differences, whereas pseudo-changes manifest as global or region-wide radiometric offsets. Aggressive alignment or normalization that removes all appearance discrepancies inevitably erodes discriminative change cues, whereas conservative correction leaves residual drift that still triggers false alarms. Domain-adaptation methods further target cross-dataset transfer rather than disentangling systematic drift from local structural change [18–21]. Effective pseudo-change suppression therefore requires separating global radiometric shifts from localized semantic differences rather than eliminating all bitemporal discrepancies. Our bias-correction module estimates channel-wise affine parameters from global bitemporal statistics and applies them at the feature front end, suppressing systematic drift while preserving local differences for the expert branches.

2.3. Mixture-of-Experts Models

Mixture-of-experts (MoE) models route inputs through multiple expert branches via a learned gating network, enabling input-dependent computation. From early gated

expert systems [25] through sparse gating [26] to Switch Transformer [27] and V-MoE [28], conditional computation has become a general framework for processing heterogeneous inputs. In remote sensing, sensor diversity has further motivated expert routing across modalities [29]. RingMoE models optical, SAR, and multispectral features with modality-specific and shared experts [30], and M^{2}CD adapts experts to optical–SAR distributional differences for change detection [31].

Existing remote-sensing MoE methods mainly organize expert specialization around sensor or modality differences, whereas the heterogeneity in bitemporal optical building change detection arises within the same image pair. Genuine building changes are primarily expressed by alterations in contours, boundaries, and local spatial structure, while cross-temporal brightness, shadow, and texture variations mainly affect appearance responses; the two types of discrepancy may also overlap spatially. To model these heterogeneous sources of difference, SARE-CD assigns persistent functional roles to the Structure, Appearance, and Adaptive Experts for geometric structural, local appearance-related, and residual composite differences, respectively, and fuses them on a shared bias-corrected change representation. Because all three experts are evaluated in every forward pass and the router continuously adjusts their spatial contributions, SARE-CD is formulated as a dense, pixel-wise, soft-gated heterogeneous MoE. This formulation differs from top-K sparse conditional computation and from selective-kernel-style fusion [32] or generic attention fusion, which typically reweight parallel transformations or existing feature responses.

3. Proposed Method

3.1. Overall Architecture

SARE-CD comprises a shared Siamese encoder, a bias-correction module, a heterogeneous expert-routing module, and a multiscale decoder (Figure 2).

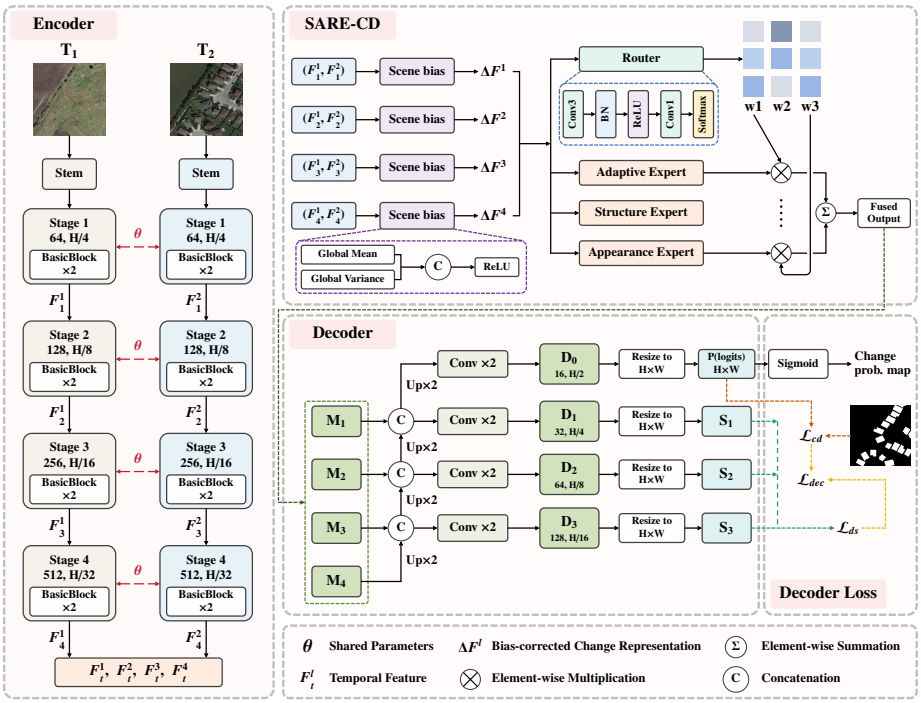

Figure 2. Overall architecture of SARE-CD. The shared Siamese encoder extracts features at four scales. Each scale undergoes bias correction and pixel-wise expert routing before the multiscale decoder produces the change probability map.

For a bitemporal image pair I_1 and I_2, a shared-weight Siamese encoder E_θ built on ResNet-18 [33] extracts features at four scales:

$$\left\{F_t^l\right\}_{l=1}^{4} = E_\theta(I_t), \quad t \in \{1,2\} \tag{1}$$

Here, $F_t^l \in \mathbb{R}^{C_l \times H_l \times W_l}$ denotes the feature map of temporal image t at encoder stage l. The four stages have dimensions $(64, H/4, W/4)$, $(128, H/8, W/8)$, $(256, H/16, W/16)$, and $(512, H/32, W/32)$, respectively. Features at all four scales are retained and passed to change-representation construction and expert modeling at their corresponding spatial levels.

At each scale, the bitemporal features form an initial change representation, ΔF_l, which the bias-correction module then transforms into $\Delta F_l'$. The module uses the global channel statistics of ΔF_l to derive spatially shared affine parameters. These parameters suppress systematic channel shifts caused by illumination and sensor calibration while preserving local differences through a residual formulation. The corrected representation, $\Delta F_l'$, enters the Adaptive, Structure, and Appearance Experts. They model composite, geometric, and local appearance differences, respectively. The pixel-wise router predicts location-specific expert weights and fuses the outputs into M_l. Bias correction and heterogeneous expert modeling operate at different statistical levels: the former adjusts scene-level channel shifts, whereas the latter models the spatially heterogeneous local differences that remain after correction. The decoder progressively aggregates the four fused feature maps, M_l, to produce the final prediction P. The overall feature flow is:

$$(I_1, I_2) \to \left\{F_1^l, F_2^l\right\}_{l=1}^{4} \to \{\Delta F_l'\}_{l=1}^{4} \to \{M_l\}_{l=1}^{4} \to P \tag{2}$$

The multiscale decoder starts from M_4 and progressively restores the spatial resolution while successively fusing M_3, M_2, and M_1. Three auxiliary predictions are produced during decoding: $S_3 \in \mathbb{R}^{H/16 \times W/16}$, $S_2 \in \mathbb{R}^{H/8 \times W/8}$, and $S_1 \in \mathbb{R}^{H/4 \times W/4}$. Before the deep-supervision loss is computed, all three outputs are resized to the input resolution $H \times W$.

3.2. Bias-Corrected Change Representation

Differences in illumination, season, and sensor state can shift bitemporal feature statistics without corresponding to genuine surface changes. We first construct an explicit change feature and then correct it with scene-level channel statistics to reduce this interference.

At scale l, the two temporal feature maps and their absolute difference are concatenated along the channel dimension. A projection then maps them into a unified C-dimensional feature space:

$$\Delta F_l = \Phi_l\Big(\mathrm{Concat}\Big[F_1^l, F_2^l, \Big|F_1^l - F_2^l\Big|\Big]\Big) \tag{3}$$

where $\Phi_l(\cdot)$ comprises a 1×1 convolution, batch normalization, and a ReLU activation. We set $C = 128$. The resulting ΔF_l jointly encodes the temporal features and their explicit difference and serves as the input to bias correction.

We then compute the channel-wise mean and variance of ΔF_l over the spatial dimensions:

$$s_l = \mathrm{Concat}[\mathrm{Mean}_{h,w}(\Delta F_l), \mathrm{Var}_{h,w}(\Delta F_l)] \tag{4}$$

A parameter-generation network maps the statistical descriptor s_l to the channel-wise scale γ_l and shift β_l:

$$[\gamma_l, \beta_l] = g_l(s_l), \qquad g_l : 2C \to C/4 \to 2C \tag{5}$$

The module then applies a residual affine transformation:

$$\Delta F_l' = (1 + \gamma_l) \odot \Delta F_l + \beta_l \tag{6}$$

The statistical descriptor $s_l \in \mathbb{R}^{2C}$ is mapped to channel-wise affine parameters by a bottleneck MLP with dimensions $2C \to C/4 \to 2C$; for $C = 128$, the corresponding dimensions are $256 \to 32 \to 256$. The hidden layer uses ReLU, and no normalization or dropout is applied inside g_l. Separate generators are used at the four scales. The output is split channel-wise into γ_l and β_l, which are spatially shared and applied to ΔF_l through the residual affine transformation in Equation 6. The final linear layer is zero-initialized, so $\gamma_l = \beta_l = 0$ at the beginning of training and the correction branch initially behaves as an identity mapping.

3.3. Structure–Appearance Heterogeneous Expert Modeling

We construct three experts to model complementary difference patterns. The Adaptive Expert handles composite differences, the Structure Expert handles geometric differences, and the Appearance Expert handles local appearance differences. A pixel-wise router fuses their responses. All three experts receive $\Delta F'_l$ as input, and Figure 3 shows their architectures.

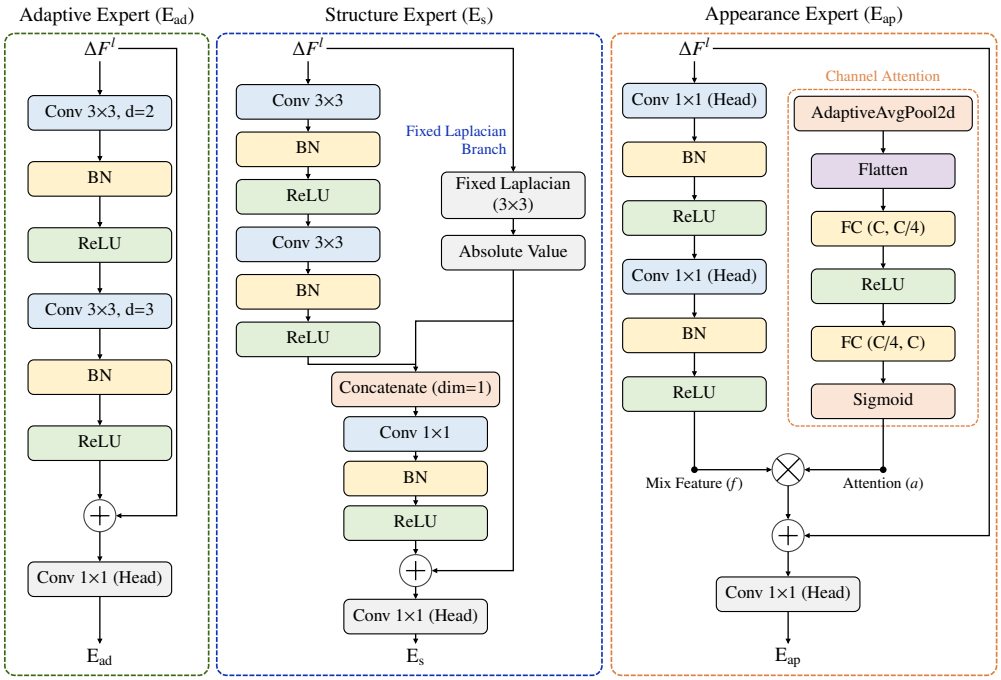

Figure 3. Architectures of the Structure, Adaptive, and Appearance Experts. The experts receive the same bias-corrected change representation and extract complementary responses through distinct processing paths.

3.3.1. Adaptive Expert

The Adaptive Expert models complex differences that the structural and appearance priors cannot adequately describe. It stacks two 3×3 dilated convolutions [34] with dilation rates of 2 and 3. This design enlarges the receptive field without reducing spatial resolution. A residual connection combines contextual information with the original change feature:

$$E_{ad}^l = H_{ad}\big(C_{d=3}\big(C_{d=2}(\Delta F_l')\big) + \Delta F_l'\big) \tag{7}$$

where H_{ad} denotes the output projection. This expert assumes no predefined difference category and captures composite patterns not captured by the Structure or Appearance Expert.

3.3.2. Structure Expert

The Structure Expert enhances responses to building boundaries, contours, and local geometric changes. It contains a learnable structural branch and a fixed Laplacian branch. The fixed branch uses the following convolution kernel:

$$K_{lap} = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix} \tag{8}$$

The fixed branch applies depthwise convolution and takes the absolute value to obtain high-frequency edge responses. In parallel, the learnable branch extracts data-dependent local structural features. The outputs are concatenated, fused, and added to the input through a residual connection:

$$E_s^l = H_s\Big(\Psi_s\Big(\text{Concat}\Big[C_s(\Delta F_l'), \Big|K_{lap} *_{dw} \Delta F_l'\Big|\Big]\Big) + \Delta F_l'\Big) \tag{9}$$

where $C_s(\cdot)$ denotes the learnable structural branch, $\Psi_s(\cdot)$ denotes feature fusion, $*_{dw}$ denotes depthwise convolution, and H_s denotes the output projection.

3.3.3. Appearance Expert

The Appearance Expert models responses induced by brightness, shadow, and texture variations. Its main branch uses two 1×1 convolutions for channel mixing and produces U_l. Channel attention [35] then estimates the importance of each channel:

$$a_l = \sigma(\text{MLP}(\text{GAP}(U_l))) \tag{10}$$

The MLP changes the channel dimension as $C \to \frac{C}{4} \to C$, and $\sigma(\cdot)$ denotes the sigmoid activation. The resulting weights recalibrate the main-branch features, which are then fused with the input through a residual connection:

$$E_{ap}^l = H_{ap}\big(U_l \odot a_l + \Delta F_l'\big) \tag{11}$$

Channel recalibration emphasizes appearance-related responses without every local difference being folded into the structural representation.

3.3.4. Pixel-Wise Routing and Expert Fusion

The dominant difference pattern varies across spatial locations. For sample b, scale l, and spatial position p, the router predicts three continuous weights in the implementation order Adaptive, Structure, and Appearance:

$$\mathbf{w}_b^l(p) = \text{Softmax}\left(\frac{R_l(\Delta F_b'^l)(p)}{\tau}\right) = \left[w_{b,ad}^l(p), w_{b,s}^l(p), w_{b,ap}^l(p)\right] \tag{12}$$

Here, $R_l(\cdot)$ consists of a 3×3 convolution, batch normalization, ReLU, and a 1×1 convolution. The temperature $\tau > 0$ controls the concentration of the routing distribution: $\tau < 1$ produces sharper expert weights and stronger local preference, whereas $\tau > 1$ yields smoother mixtures across the three experts. We use $\tau = 1.0$ in all main experiments and evaluate its sensitivity separately in Section 4.5.7. The three weights are nonnegative and sum to one at every location.

$$M_b^l(p) = \sum_{k \in \{ad,s,ap\}} w_{b,k}^l(p)\, E_{k,b}^l(p) \tag{13}$$

All three experts are evaluated in every forward pass.

3.4. Training Objective and Routing-Weight Visualization

The training objective jointly optimizes change prediction and expert routing. The total loss is:

$$\mathcal{L} = \mathcal{L}_{cd} + \lambda_{ds}\mathcal{L}_{ds} + \lambda_{\text{ent}}\mathcal{L}_{\text{ent}} + \lambda_{bal}\mathcal{L}_{bal} \tag{14}$$

where $\mathcal{L}_{cd}$ is the main change-detection loss and $\mathcal{L}_{ds}$ is the deep-supervision loss. The terms $\mathcal{L}_{\text{ent}}$ and $\mathcal{L}_{bal}$ constrain local routing certainty and batch-level expert usage, respectively.

3.4.1. Change Detection Loss and Deep Supervision

We optimize the final change prediction with the sum of weighted binary cross-entropy and Dice loss [36]:

$$\mathcal{L}_{cd} = \mathcal{L}_{BCE} + \mathcal{L}_{Dice} \tag{15}$$

The positive-class weight in the binary cross-entropy term is computed from the current batch's negative-to-positive pixel ratio and clipped to $[1, 15]$. The auxiliary predictions S_3, S_2, and S_1 are produced at $H/16 \times W/16$, $H/8 \times W/8$, and $H/4 \times W/4$, respectively. Let $U(\cdot)$ denote resizing to the input resolution $H \times W$; the deep-supervision loss is then computed after scale alignment:

$$\mathcal{L}_{ds} = 0.2\mathcal{L}_{cd}(U(S_3), Y) + 0.3\mathcal{L}_{cd}(U(S_2), Y) + 0.4\mathcal{L}_{cd}(U(S_1), Y) \tag{16}$$

The scale-specific coefficients 0.2, 0.3, and 0.4 control the relative importance of the three auxiliary predictions, while $\lambda_{\text{ds}} = 0.4$ in Eq. (14) controls the overall contribution of the deep-supervision loss to the total objective. The effective weights are therefore 0.08, 0.12, and 0.16. These coefficients are empirical tuning factors and are not required to sum to 1.0; their sum is 0.9 simply because no auxiliary prediction was assigned a higher relative weight during validation-based selection.

3.4.2. Routing Entropy and Load-Balancing Constraints

Routing entropy and load balancing constrain the soft assignments at different statistical levels. The entropy term acts directly on the three-expert distribution at each pixel and regulates the concentration of local routing, whereas load balancing measures average expert usage at each scale and limits persistent bias toward a small subset of branches.

$$\mathcal{L}_{\text{ent}} = -\frac{1}{B\sum_{l=1}^{4} H_l W_l} \sum_{l=1}^{4} \sum_{b=1}^{B} \sum_{p=1}^{H_l W_l} \sum_{k=1}^{3} w_{b,k}^{l}(p) \log\Big(w_{b,k}^{l}(p) + \varepsilon\Big) \tag{17}$$

Equation (17) uses the Shannon entropy in nats. The expert dimension is summed first, followed by averaging over batch and spatial positions, with each scale weighted by $H_l W_l$. The constant ε is used only for numerical stability in the logarithm and does not renormalize the Softmax probabilities. With three experts, the maximum single-pixel entropy is $\ln 3$. At scale l, the mean usage of expert k is $\bar{w}_{l,k} = \frac{1}{BH_l W_l}\sum_{b=1}^{B}\sum_{p=1}^{H_l W_l} w_{b,k}^{l}(p)$. The load-balancing term then measures its deviation from the uniform reference $1/3$.

$$\mathcal{L}_{bal} = \frac{1}{\sum_{l=1}^{4} H_l W_l} \sum_{l=1}^{4} H_l W_l \sum_{k=1}^{3} \left(\bar{w}_{l,k} - \frac{1}{3}\right)^2 \tag{18}$$

Accordingly, $\mathcal{L}_{\text{ent}}$ regulates the concentration of the local routing distribution, whereas $\mathcal{L}_{bal}$ regulates the balance of global expert usage.

3.4.3. Routing-Weight Visualization

For visualization, the dominant expert for sample b, scale l, and location p is defined by the largest of the three routing weights:

$$D_b^l(p) = \underset{k\in\{ad,s,ap\}}{\arg\max}\ w_{b,k}^l(p) \tag{19}$$

The dominant-expert map is used only to discretize the continuous routing weights for visualization; training and prediction always use the continuous weights in Equation 13, and the later spatial analyses likewise retain the continuous routing values.

4. Experiments

We evaluate SARE-CD through datasets and implementation details, baseline comparisons, quantitative and qualitative results, complexity analysis, and ablation studies. These experiments assess cross-dataset performance and the contributions of the proposed representation, expert specialization, and training constraints.

4.1. Datasets and Experimental Settings

We evaluate SARE-CD on three public benchmarks, LEVIR-CD+ [22], S2Looking [23], and SYSU-CD [24], summarized in Table 1. They differ in spatial resolution, viewing geometry, change-target scale, and appearance interference: LEVIR-CD+ contains high-resolution urban buildings with relatively regular contours, S2Looking is acquired under side-looking geometry with pronounced shadow and viewpoint shifts, and SYSU-CD contains complex backgrounds whose textures can resemble building changes. All images are cropped to 256 × 256 for training and testing, consistent with most comparison methods.

Table 1. Experimental datasets and input settings.

Dataset	Training/Validation/Test	Input Size	Main Challenges
LEVIR-CD+	8144/2048/5568	256 × 256	High-resolution urban buildings; high requirement for boundary integrity
S2Looking	56,000/8000/16,000	256 × 256	Side-looking imaging; pronounced shadow and viewpoint shifts
SYSU-CD	12,000/4000/4000	256 × 256	Complex urban scenes; strong background interference

We report recall, precision, overall accuracy (OA), F1, and intersection over union (IoU). Because unchanged pixels dominate change-detection images, class imbalance inflates OA; we therefore treat F1 and IoU as the primary metrics and use recall and precision to interpret false-negative and false-positive behavior.

All three datasets use the same model configuration. An ImageNet-pretrained ResNet-18 [33] serves as the shared-weight Siamese encoder, and each encoder scale is projected to $C = 128$ channels before bias correction and expert routing. The projection width is selected on the validation set and then fixed for all experiments. For network optimization, the model employs the AdamW optimizer [37]. The initial learning rate is set to 1.5×10^{-4} for the ImageNet-pretrained encoder and 3×10^{-4} for the newly initialized modules, with a weight decay of 1×10^{-4}. The learning rate schedule incorporates a linear warm-up for the first 10 epochs, followed by a cosine annealing strategy. The maximum number of training epochs is 150. The batch size is 32. Model selection is based on validation F1, with

an early-stopping patience of 50 epochs, and the checkpoint with the highest validation F1 is used for evaluation.

The loss weights are fixed to $\lambda_{ds} = 0.4$, $\lambda_{\text{ent}} = 0.05$, and $\lambda_{bal} = 0.10$, and the routing temperature is $\tau = 1.0$. A single binarization threshold of 0.50 is used for all datasets, models, ablations, and perturbation experiments. Regarding data augmentation, random horizontal flipping ($p = 0.5$), random vertical flipping ($p = 0.5$), and random rotation by multiples of 90° are synchronously applied to I_1, I_2, and the ground-truth labels. Each configuration is trained independently with random seeds 42, 43, and 44, and repeated-run results are reported as mean ± standard deviation where applicable. Experiments are conducted in PyTorch 2.4.0 with CUDA 12.4 on one NVIDIA GeForce RTX 5090 32 GB GPU using FP32.

It is important to emphasize that the selection of hyperparameters (e.g., the projection width $C = 128$ and the routing loss weights $\lambda_{\text{ent}} = 0.05, \lambda_{bal} = 0.10$) was performed strictly and independently on the official 4000-image validation set of SYSU-CD. The test set remained completely independent and was used exclusively for final performance evaluation after the hyperparameters were frozen; it was never involved in any parameter tuning or checkpoint selection process.

4.2. Comparison Methods

The comparison includes eight representative change-detection methods covering convolutional, Transformer, state-space, and semantic-prior architectures. FC-EF, FC-SiamDiff, and FC-SiamConc represent input-level fusion, feature differencing, and feature concatenation, respectively [6]; SNUNet-CD represents dense multiscale convolutional fusion [7]; BIT and ChangeFormer represent Transformer-based temporal modeling [10,11]; ChangeMamba represents state-space modeling [13]; and SA-CDNet is included as a recent semantic-prior method [38]. All reproduced methods use the same input size and evaluation protocol as SARE-CD while retaining their original architectures and official pretraining settings where applicable.

4.3. Comparison with State-of-the-Art Methods

We compare SARE-CD with representative methods on LEVIR-CD+, S2Looking, and SYSU-CD. SARE-CD differs by avoiding a single shared discrimination path for all temporal differences. Instead, bias correction, heterogeneous experts, and pixel-wise routing separately model structural, appearance-related, and residual composite differences.

We first compare predictions on LEVIR-CD+, where boundary integrity most readily exposes a model's structural modeling ability (Figure 4). Building contours are relatively distinct, so the main challenge is preserving roof boundaries, small buildings, and separation between adjacent structures. Several baselines show discontinuous boundaries or miss small targets. SARE-CD produces more continuous regions and better-constrained boundaries, which is consistent with the intended role of the Structure Expert in enhancing local geometric cues.

Table 2 shows that SARE-CD achieves an F1 score of 91.58% and an IoU of 84.47% on LEVIR-CD+. Both are the highest values among the compared methods.

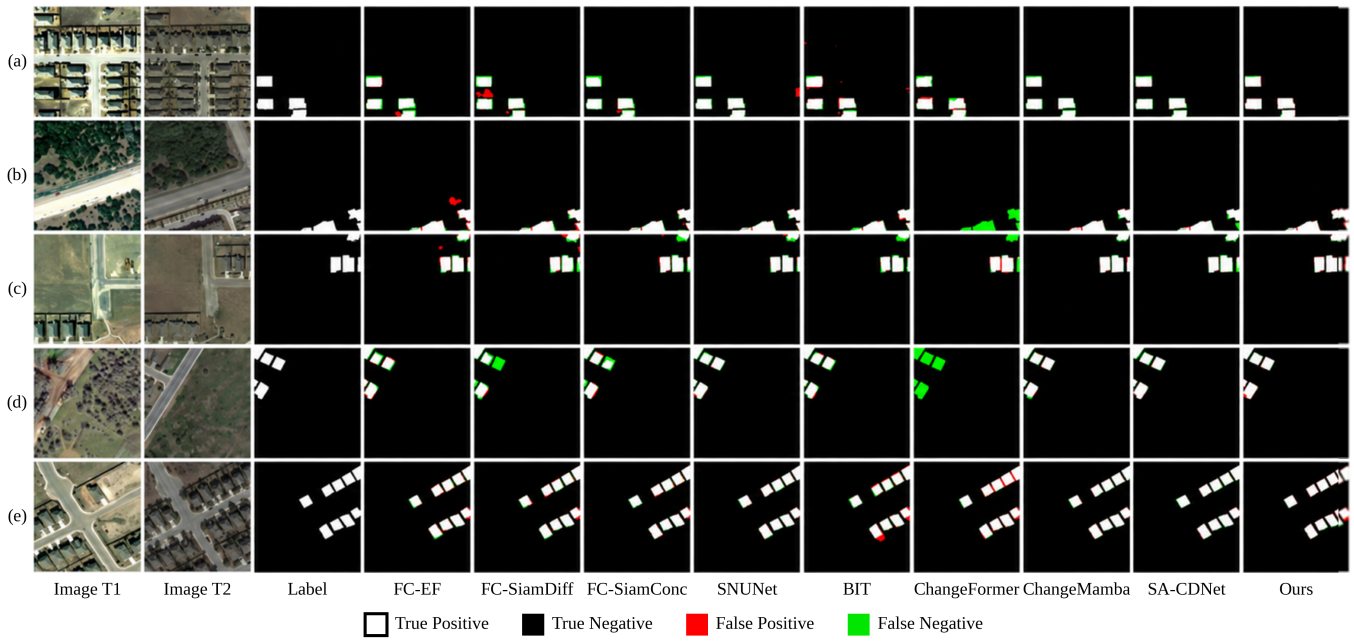

Figure 4. Visual comparison on LEVIR-CD+. (**a**) Roof expansions in complex residential areas; (**b**) subtle changes along roads and vegetation; (**c**) new constructions in open areas; (**d**) changes in scattered buildings; (**e**) building changes in densely planned areas. White, black, red, and green denote true positives, true negatives, false positives, and false negatives, respectively. The examples highlight roof-boundary preservation, small-building detection, and the separation of adjacent structures.

Table 2. Quantitative comparison on LEVIR-CD+ (mean ± std, %). The best value in each metric is marked in bold.

Method	Recall	Precision	OA	F1	IoU
FC-EF	81.43 ± 0.45	85.97 ± 0.43	98.57 ± 0.05	83.64 ± 0.31	71.88 ± 0.46
FC-SiamDiff	79.16 ± 0.41	87.98 ± 0.39	98.58 ± 0.04	83.34 ± 0.28	71.43 ± 0.41
FC-SiamConc	82.17 ± 0.38	89.36 ± 0.36	98.76 ± 0.04	85.62 ± 0.26	74.85 ± 0.40
SNUNet-CD	85.25 ± 0.35	87.18 ± 0.33	98.77 ± 0.04	86.20 ± 0.24	75.75 ± 0.37
BIT	80.52 ± 0.39	88.42 ± 0.37	98.65 ± 0.04	84.28 ± 0.27	72.84 ± 0.40
ChangeFormer	86.62 ± 0.28	91.60 ± 0.26	99.04 ± 0.03	89.04 ± 0.19	80.24 ± 0.31
ChangeMamba	86.02 ± 0.30	87.89 ± 0.28	98.84 ± 0.03	86.95 ± 0.20	76.91 ± 0.32
SA-CDNet	89.29 ± 0.22	**91.87 ± 0.20**	99.16 ± 0.02	90.56 ± 0.17	82.75 ± 0.28
SARE-CD (Ours)	**92.07 ± 0.16**	91.10 ± 0.15	**99.24 ± 0.02**	**91.58 ± 0.07**	**84.47 ± 0.19**

S2Looking presents stronger cross-temporal differences in roof appearance, shadow, and local texture than LEVIR-CD+ because of its side-looking geometry. These differences make change discrimination more difficult. Figure 5 compares the methods' false-positive and false-negative distributions within change regions and nearby interference regions.

Table 3 shows substantial variation in the balance between recall and precision on S2Looking. SARE-CD achieves the highest F1 score, 65.78%, and the highest IoU, 49.01%. These results indicate the strongest overall precision–recall balance among the compared methods in complex side-looking scenes.

In SYSU-CD, building changes coexist with roads, bare soil, shadows, and roof textures, which makes a single shared enhancement path more susceptible to precision–recall imbalance. Figure 6 shows that SARE-CD more consistently separates genuine change boundaries from local appearance perturbations. The specialized experts reduce extensive false positives and improve the shape consistency of detected regions, which we attribute to the separate modeling of geometric and appearance differences by the expert branches.

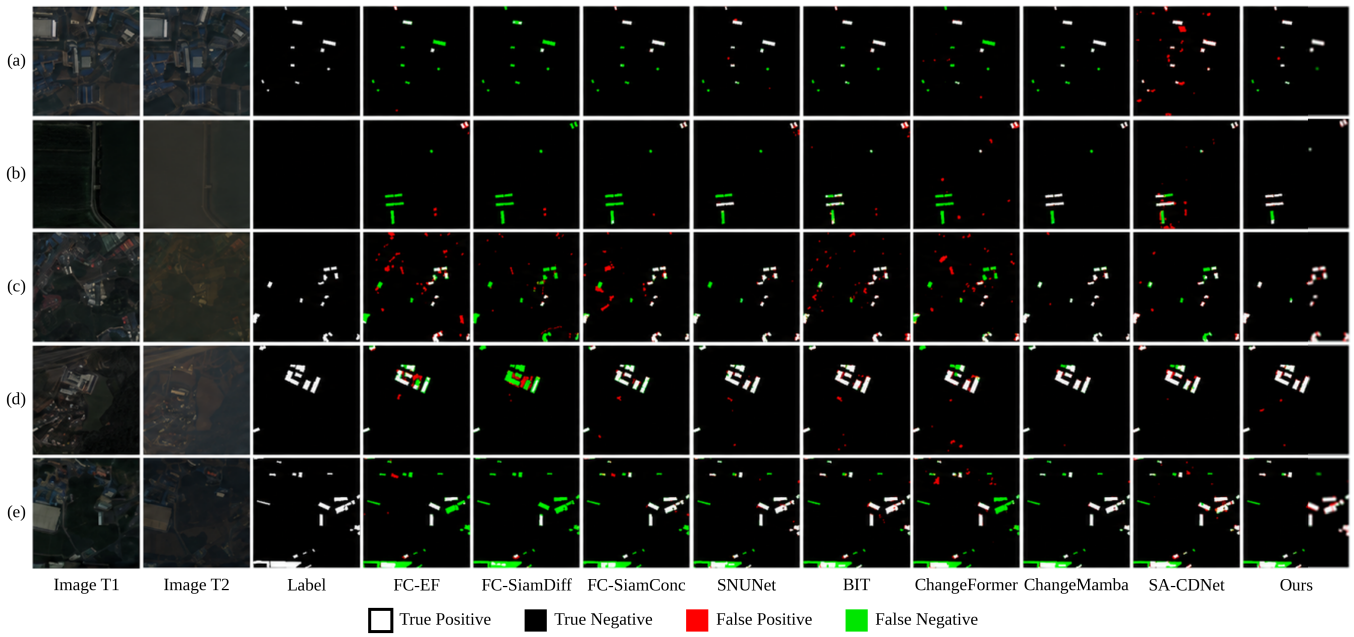

Figure 5. Visual comparison on S2Looking. (**a**) Small building changes in dense areas; (**b**) linear changes along agricultural facilities; (**c**) complex changes in irregular building areas; (**d**) structural changes in large industrial complexes; (**e**) newly added structures in industrial zones. White, black, red, and green denote true positives, true negatives, false positives, and false negatives, respectively. The examples highlight detection under side-looking geometry, where shadow displacement and roof-appearance drift across time are pronounced.

Table 3. Quantitative comparison on S2Looking (mean ± std, %). The best value in each metric is marked in bold.

Method	Recall	Precision	OA	F1	IoU
FC-EF	36.60 ± 0.78	46.92 ± 0.74	98.73 ± 0.05	41.12 ± 0.62	25.88 ± 0.49
FC-SiamDiff	47.97 ± 0.71	55.96 ± 0.68	98.91 ± 0.04	51.66 ± 0.55	34.82 ± 0.50
FC-SiamConc	48.13 ± 0.69	55.09 ± 0.66	98.89 ± 0.04	51.37 ± 0.53	34.56 ± 0.49
SNUNet-CD	62.09 ± 0.54	64.83 ± 0.50	99.13 ± 0.03	63.43 ± 0.41	46.45 ± 0.44
BIT	59.06 ± 0.58	63.03 ± 0.54	99.08 ± 0.03	60.98 ± 0.45	43.87 ± 0.43
ChangeFormer	**78.45 ± 0.61**	38.26 ± 0.58	98.20 ± 0.06	51.44 ± 0.48	34.62 ± 0.45
ChangeMamba	61.30 ± 0.49	63.67 ± 0.46	99.10 ± 0.04	62.47 ± 0.37	45.42 ± 0.40
SA-CDNet	45.37 ± 0.52	**73.42 ± 0.49**	**99.14 ± 0.03**	56.09 ± 0.40	38.97 ± 0.42
SARE-CD (Ours)	70.39 ± 0.31	61.73 ± 0.30	99.11 ± 0.03	**65.78 ± 0.67**	**49.01 ± 0.27**

As shown in Table 4, SARE-CD achieves an F1 score of 84.46% and an IoU of 73.10%. Both are the highest among the compared methods. The gains over second-ranked ChangeMamba are 1.61 and 2.38 percentage points, respectively. These results indicate strong overall detection performance in complex urban backgrounds, suggesting that the heterogeneous expert design is effective in such scenes.

Table 4. Quantitative comparison on SYSU-CD (mean ± std, %). The best value in each metric is marked in bold.

Method	Recall	Precision	OA	F1	IoU
FC-EF	76.94 ± 0.39	75.58 ± 0.37	88.70 ± 0.12	76.25 ± 0.30	61.62 ± 0.37
FC-SiamDiff	80.54 ± 0.36	73.11 ± 0.35	88.43 ± 0.13	76.65 ± 0.29	62.13 ± 0.36
FC-SiamConc	76.02 ± 0.41	72.70 ± 0.39	87.62 ± 0.14	74.32 ± 0.31	59.14 ± 0.38
SNUNet-CD	73.89 ± 0.42	74.76 ± 0.40	87.96 ± 0.11	74.32 ± 0.32	59.14 ± 0.39
BIT	76.35 ± 0.35	78.27 ± 0.34	89.43 ± 0.11	77.30 ± 0.27	62.99 ± 0.34
ChangeFormer	73.11 ± 0.33	83.50 ± 0.31	90.25 ± 0.10	77.96 ± 0.25	63.88 ± 0.32
ChangeMamba	77.23 ± 0.27	**89.35 ± 0.25**	92.46 ± 0.08	82.85 ± 0.21	70.72 ± 0.27
SA-CDNet	81.94 ± 0.29	81.76 ± 0.27	91.43 ± 0.09	81.85 ± 0.22	69.27 ± 0.28
SARE-CD (Ours)	**84.17 ± 0.20**	84.76 ± 0.18	**92.70 ± 0.07**	**84.46 ± 0.08**	**73.10 ± 0.22**

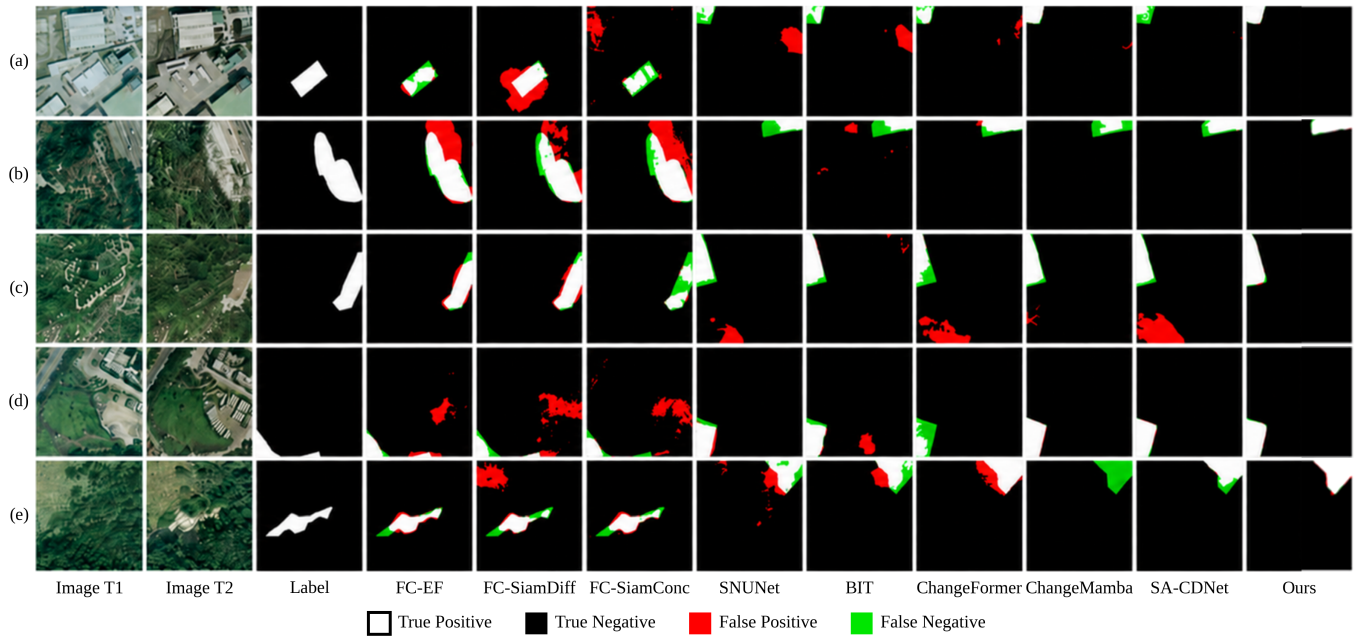

Figure 6. Visual comparison on SYSU-CD. (**a**) Expansion of large industrial buildings; (**b**) boundary changes of large irregular sites; (**c**) linear structural changes near roads; (**d**) new constructions on bare soil or open areas; (**e**) building changes along complex terrain and water bodies. White, black, red, and green denote true positives, true negatives, false positives, and false negatives, respectively. The examples highlight the separation of genuine change boundaries from road, bare-soil, shadow, and roof-texture interference.

Beyond parameter count and FLOPs, we measure practical inference efficiency under the same hardware environment with batch size 1. To ensure a fair comparison of computational complexity, the parameter counts and FLOPs in Table 5 were remeasured using a unified protocol: utilizing the fvcore tool, with the input tensor fixed at $[1, 3, 256, 256]$ and a batch size of 1. The measurement encompasses the encoder, fusion module, decoder, and final output head; a fused multiply-add (FMA) operation is counted as one FLOP, and overheads such as data loading, preprocessing, and the deep-supervision heads (which are used only during training) are strictly excluded. Table 5 reports latency, throughput, and peak GPU memory together with model complexity. Latency is measured in evaluation mode with gradient computation disabled and CUDA synchronization before and after each forward pass; peak memory is recorded with `torch.cuda.max_memory_allocated()`.

SARE-CD requires 5.74 ms per bitemporal pair, corresponding to 174.22 FPS, with a peak GPU memory usage of 151.8 MB. The three-expert design introduces more latency than the simplest FC-based models, but remains faster than ChangeFormer and ChangeMamba under the same measurement protocol. Together with 15.42 M parameters and 20.41 G

FLOPs, these measurements place SARE-CD at a moderate deployment cost rather than in the ultra-lightweight regime.

Table 5. Complexity and measured inference efficiency under a unified batch-size-1 protocol. Bold indicates the proposed method.

Method	Parameters (M)	FLOPs (G)	Latency (ms/Pair)	FPS	Peak Memory (MB)
FC-EF	1.35	3.60	1.32	757.58	57.1
FC-SiamDiff	1.35	4.72	1.93	518.13	67.9
FC-SiamConc	1.54	5.31	1.88	531.91	71.6
SNUNet-CD	12.03	27.44	9.86	101.42	232.6
BIT	3.55	4.35	6.31	158.48	108.7
ChangeFormer	41.03	202.86	11.27	88.73	224.3
ChangeMamba	84.70	179.32	27.98	35.74	550.8
SA-CDNet	77.08	12.03	69.46	14.40	439.8
SARE-CD (Ours)	**15.42**	**20.41**	**5.74**	**174.22**	**151.8**

4.4. Fine-Grained Structural and Robustness Evaluation

Building-change recognition is jointly affected by contour localization, the proportion of changed pixels, and non-structural appearance differences. We therefore analyze boundary quality, sparse-change behavior, appearance-induced pseudo-change responses, and controlled perturbations separately.

4.4.1. Building-Boundary Delineation Quality

Building changes often involve alterations in roof contours and local spatial layouts, so region overlap alone does not fully describe localization quality. We therefore report boundary F1 (BF1) and boundary IoU (BIoU) [39]. BF1 uses a matching tolerance of $0.75\%D$, and BIoU uses a boundary width of $2\%D$, where D is the image diagonal. For a 256×256 image, the image diagonal $D \approx 362$ pixels, so $0.75\%D \approx 2.7$ pixels and $2\%D \approx 7.2$ pixels. ChangeMamba, SA-CDNet, and SARE-CD are evaluated with the same boundary implementation and output resolution. The boundary comparison is limited to these two competing methods because only they provide consistent boundary predictions or publicly available source code with the same output resolution; including methods without reliable boundary outputs would result in an unfair comparison. Table 6 reports the boundary quality comparison for the three datasets.

Table 6. Boundary quality comparison (mean ± std, %). The best value in each metric is marked in bold.

Dataset	Method	BF1	BIoU
LEVIR-CD+	ChangeMamba	84.21 ± 0.31	62.90 ± 0.44
	SA-CDNet	83.70 ± 0.28	62.18 ± 0.40
	SARE-CD	**90.60 ± 0.19**	**70.85 ± 0.31**
S2Looking	ChangeMamba	61.92 ± 0.42	43.85 ± 0.55
	SA-CDNet	58.40 ± 0.46	40.27 ± 0.58
	SARE-CD	**64.75 ± 0.35**	**46.68 ± 0.48**
SYSU-CD	ChangeMamba	50.66 ± 0.34	36.98 ± 0.45
	SA-CDNet	48.91 ± 0.37	35.34 ± 0.47
	SARE-CD	**52.82 ± 0.29**	**39.07 ± 0.40**

Building contours are relatively regular in LEVIR-CD+, where SARE-CD reaches 90.60% BF1 and 70.85% BIoU. The absolute boundary scores are lower on S2Looking and SYSU-CD because viewpoint variation, shadows, and complex background textures make

contour localization more difficult; nevertheless, SARE-CD achieves the highest BF1 and BIoU among the three compared methods on all datasets. Because boundary metrics still aggregate all changed pixels, the next analysis focuses specifically on sparse-change cases.

4.4.2. Sensitivity to Change-Area Proportion

For each ground-truth mask, the changed-pixel ratio is defined as

$$r = \frac{N_{change}}{N_{total}} \tag{20}$$

Samples with $0 < r < 5\%$ are treated as sparse-change cases, while all-unchanged samples with $r = 0$ are excluded. To keep the sparse-change statistics consistent with the pooled pixel-level evaluation used in the main comparison, Table 7 aggregates confusion counts over the complete SYSU-CD Small subset before computing Recall, Precision, F1, and IoU. The same sample list and evaluation procedure are used for all three methods.

Table 7. Sparse-change comparison on the Small subsets ($0 < r < 5\%$, unit: %). All metrics are computed using pooled pixel-level statistics. The best value in each metric is marked in bold.

Dataset	Method	Recall	Precision	F1	IoU
LEVIR-CD+	ChangeMamba	69.84	84.50	76.47	61.91
	SA-CDNet	**79.67**	**88.20**	**83.72**	**72.00**
	SARE-CD	78.57	79.96	79.26	65.64
S2Looking	ChangeMamba	59.26	62.00	60.60	43.47
	SA-CDNet	38.41	**72.00**	50.10	33.42
	SARE-CD	**69.69**	59.03	**63.92**	**46.97**
SYSU-CD	ChangeMamba	58.25	44.88	50.70	33.96
	SA-CDNet	52.18	**49.50**	**50.80**	**34.05**
	SARE-CD	**64.16**	35.21	45.47	29.42

Under the same definition for sparse changes ($0 < r < 5\%$), SARE-CD achieves F1 scores of 79.26% and 63.92% on the Small subsets of LEVIR-CD+ and S2Looking, respectively. Combined with the performance on SYSU-CD (45.47% F1), these results indicate that although the model maintains relatively high sensitivity to sparse changes in scenes with regular building structures, false positives in complex backgrounds remain a clear limitation for sparse-target detection accuracy. Section 5.2 further examines the routing behavior associated with these errors.

For comparison, on the LEVIR-CD+ Small subset, SA-CDNet achieves 83.72% F1 and 72.00% IoU, outperforming SARE-CD by 4.46 and 6.36 percentage points, respectively. On the SYSU-CD Small subset, ChangeMamba and SA-CDNet reach 50.70% and 50.80% F1, respectively, exceeding SARE-CD by approximately 5.2–5.3 percentage points. The lower F1 on SYSU-CD Small is primarily due to the low Precision of 35.21%, indicating that false positives in cluttered backgrounds overwhelm the true sparse changes.

4.4.3. Suppression of Appearance-Induced Pseudo-Changes

The Appearance Expert is designed to reduce responses caused by cross-temporal brightness, shadow, and texture differences without suppressing genuine structural changes. We evaluate three independently trained configurations: M_0 removes the Appearance Expert, M_1 retains the expert without channel attention (CA), and M_2 uses the complete Appearance Expert with CA.

For each dataset, let $R_0 = \{x \mid y_x = 0\}$ denote the fixed set of all ground-truth unchanged pixels. Under condition c, the mean change confidence of configuration m is

$$\overline{P}_{chg}^{(m,c)} = \frac{1}{|R_0|} \sum_{x \in R_0} p_m(y_x = 1 \mid x, I_c) \times 100\%. \tag{21}$$

A lower $\overline{P}_{chg}$ indicates weaker erroneous change confidence on unchanged pixels. Binary detection metrics are computed from the same probability maps with the fixed threshold

$$\hat{y}_x^{(m,c)} = \mathbb{I}[p_m(y_x = 1 \mid x, I_c) \geq 0.50]. \tag{22}$$

Table 8 summarizes the natural-condition appearance ablation. Across all three datasets, the mean change confidence on ground-truth unchanged pixels decreases consistently from M_0 to M_1 and then to M_2, while F1 is maintained or improved. On LEVIR-CD+, for example, M_2 reduces Recall by 0.522 percentage points relative to M_0 but increases Precision by 1.942 points, raising F1 from 90.840% to 91.580%. The synchronized change in probability response and detection metrics indicates that the lower unchanged-region confidence mainly corresponds to reduced false positives rather than a uniform suppression of all change predictions. Because the datasets differ in scene composition, viewpoint, background texture, and class distribution, the absolute magnitude of $\overline{P}_{chg}$ is not treated as a calibrated quantity for direct cross-dataset comparison. We therefore interpret this statistic primarily through the M_0–M_1–M_2 trend within the same dataset and evaluation condition.

Table 8. Natural-condition appearance ablation. Mean $\overline{P}_{chg}$ is computed over ground-truth unchanged pixels (%; lower is better). Bold values highlight the best F1 and the lowest mean $\overline{P}_{chg}$.

Dataset	Variant	Recall	Precision	F1	Mean $\overline{P}_{chg}$
LEVIR-CD+	M_0: w/o Appearance	92.592	89.153	90.840 ± 0.07	5.620
	M_1: Appearance w/o CA	92.483	89.797	91.120 ± 0.15	3.360
	M_2: Appearance + CA	92.070	91.095	**91.580 ± 0.07**	**1.900**
S2Looking	M_0: w/o Appearance	68.712	60.509	64.350 ± 0.33	46.70
	M_1: Appearance w/o CA	69.913	60.767	65.020 ± 0.28	38.85
	M_2: Appearance + CA	70.390	61.730	**65.780 ± 0.67**	**33.60**
SYSU-CD	M_0: w/o Appearance	83.123	82.997	83.060 ± 0.12	40.95
	M_1: Appearance w/o CA	83.663	83.677	83.670 ± 0.20	33.25
	M_2: Appearance + CA	84.170	84.760	**84.460 ± 0.08**	**27.55**

Starting from the natural-image setting, we further perturb only T_2 with multiplicative brightness change and soft shadow at severity S3 while keeping T_1, the ground truth, and the unchanged-pixel set R_0 fixed. Recall, Precision, F1, and mean $\overline{P}_{chg}$ in Tables 9 and 10 are computed from the same predictions, allowing the responses of the Appearance Expert and channel attention to be compared under controlled appearance changes.

The same overall tendency is preserved under brightness and soft-shadow perturbations. On SYSU-CD, F1 increases from 82.550% for M_0 to 83.550% for M_2 under brightness S3, while mean $\overline{P}_{chg}$ decreases from 45.60% to 30.10%; under soft shadow, the corresponding values change from 82.200%/43.55% to 83.730%/28.85%. On S2Looking, M_2 shows a small Recall decrease, but the accompanying Precision gain offsets this loss and yields a higher F1 than M_0. These results indicate that the Appearance Expert and channel attention mainly reshape the false-positive/false-negative trade-off rather than uniformly suppressing change responses. The fixed-S3 experiments isolate the contribution of the Appearance branch, whereas the stability of the complete model across continuous perturbation levels is evaluated next.

Table 9. Appearance ablation under brightness perturbation (S3). Mean $\overline{P}_{chg}$ is computed over ground-truth unchanged pixels (%; lower is better). Bold values highlight the best F1 and the lowest mean $\overline{P}_{chg}$.

Dataset	Variant	Recall	Precision	F1	Mean $\overline{P}_{chg}$
LEVIR-CD+	M_0	92.544	88.353	90.400 ± 0.09	8.35
	M_1	92.416	88.700	90.520 ± 0.10	4.62
	M_2	92.573	89.017	**90.760 ± 0.11**	**2.35**
S2Looking	M_0	68.230	61.247	64.550 ± 0.57	49.90
	M_1	67.442	62.394	64.820 ± 0.51	43.10
	M_2	66.857	63.092	**64.920 ± 0.58**	**36.20**
SYSU-CD	M_0	83.441	81.678	82.550 ± 0.14	45.60
	M_1	83.127	82.376	82.750 ± 0.14	36.85
	M_2	82.620	84.501	**83.550 ± 0.11**	**30.10**

Table 10. Appearance ablation under soft-shadow perturbation (S3). Mean $\overline{P}_{chg}$ is computed over ground-truth unchanged pixels (%; lower is better). Bold values highlight the best F1 and the lowest mean $\overline{P}_{chg}$.

Dataset	Variant	Recall	Precision	F1	Mean $\overline{P}_{chg}$
LEVIR-CD+	M_0	92.316	88.370	90.300 ± 0.11	7.22
	M_1	92.215	88.694	90.420 ± 0.12	4.05
	M_2	92.277	89.583	**90.910 ± 0.07**	**2.18**
S2Looking	M_0	68.161	61.122	64.450 ± 0.58	49.95
	M_1	67.308	62.324	64.720 ± 0.49	41.30
	M_2	66.808	63.439	**65.080 ± 0.56**	**34.90**
SYSU-CD	M_0	82.416	81.985	82.200 ± 0.10	43.55
	M_1	82.367	82.433	82.400 ± 0.12	35.20
	M_2	81.712	85.850	**83.730 ± 0.07**	**28.85**

4.4.4. Robustness to Cross-Temporal Appearance and Geometric Variations

Robustness is evaluated by keeping T_1 and the ground truth fixed and perturbing only T_2. Brightness uses the multiplicative transformation $I_2' = \mathrm{clip}[(1 \pm b)I_2]$ with $b \in \{0.10, 0.20, 0.30, 0.40, 0.50\}$; hue is cyclically shifted in HSV space by $h \in \{0.02, 0.04, 0.06, 0.08, 0.10\}$, while saturation and value are kept unchanged; soft shadow uses $I_2' = \mathrm{clip}[I_2(1 - aM)]$ with $a \in \{0.10, 0.20, 0.30, 0.40, 0.50\}$; Gaussian blur uses $\sigma \in \{1, 2, 3, 4, 6\}$; and misregistration shifts the image by $d \in \{1, 2, 3, 4, 5\}$ pixels along the four cardinal directions. Severity 0 denotes the unperturbed input. The five severity levels are controlled stress tests spanning mild to strong photometric or geometric deviations while preserving identifiable scene content.

The absolute F1 curves reveal a clear difference between photometric variation and spatial-detail degradation. Brightness, hue, and soft shadow cause relatively gradual performance changes within the tested range, whereas Gaussian blur produces a much steeper decline as severity increases; on LEVIR-CD+, F1 falls to 22.30% at S5 blur. The effect of spatial misregistration lies between these cases and varies across datasets.

The corresponding F1 retention is

$$R_{F1}(s) = \frac{F1(s)}{F1(0)} \times 100\%. \tag{23}$$

Table 11 and Figure 7 describe the same perturbation experiments in absolute and normalized form, respectively. At S5, brightness retention is 97.61%, 96.72%, and 97.35% on LEVIR-CD+, S2Looking, and SYSU-CD; hue retention is 99.17%, 98.39%, and 98.72%; and

soft-shadow retention is 98.38%, 97.57%, and 97.93%. In contrast, Gaussian-blur retention drops to 24.35%, 64.12%, and 67.04%, while misregistration retention is 81.68%, 91.58%, and 95.00%.

It is important to note that the impact of brightness perturbations is highly dependent on the protocol definition. We tested both multiplicative brightness (proportional gain changes) and additive brightness (global fixed offsets). At the maximum perturbation level (Severity 5), SARE-CD demonstrates high stability under multiplicative brightness changes, retaining 97.61%, 96.72%, and 97.35% of its original F1 score on LEVIR-CD+, S2Looking, and SYSU-CD, respectively. However, under extreme additive global brightness offsets, where a large number of pixels are clipped, the F1 retention rates drop to 70.5%, 29.3%, and 88.7%, respectively. This indicates that while the model is highly robust to proportional variations in relative illumination or sensor gain, it remains sensitive to extreme absolute brightness offsets.

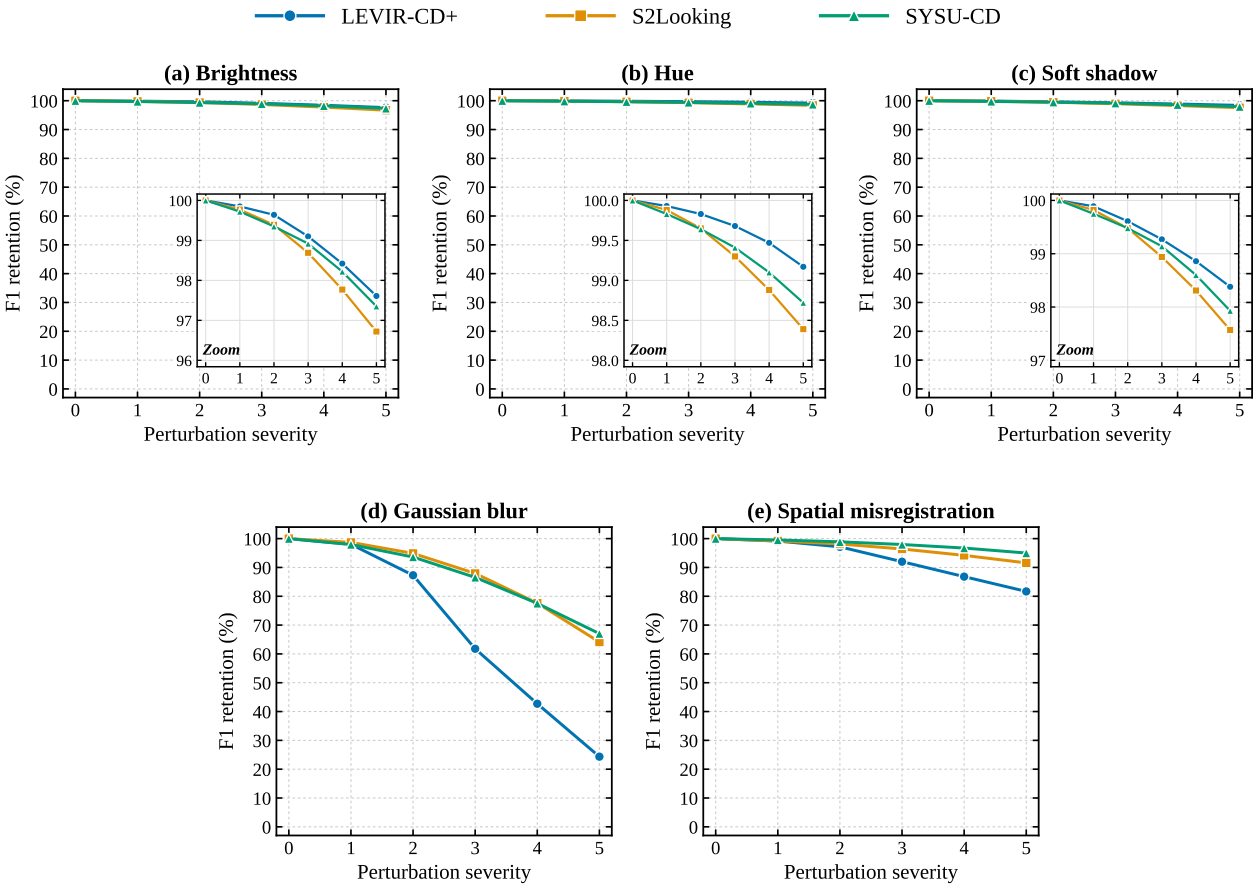

Figure 7. F1 retention of SARE-CD under five controlled perturbations applied to T_2 only: (**a**) brightness, (**b**) hue, and (**c**) soft shadow in the top row and (**d**) Gaussian blur and (**e**) spatial misregistration in the bottom row. Severity 0 denotes the unperturbed image. All panels use the same 0–100% vertical scale.

Table 11. Absolute F1 under controlled perturbations (mean ± std, %).

Dataset	Perturbation	S0	S1	S2	S3	S4	S5
LEVIR-CD+	Brightness	91.58 ± 0.07	91.44 ± 0.09	91.25 ± 0.08	90.76 ± 0.11	90.13 ± 0.07	89.39 ± 0.10
	Hue	91.58 ± 0.07	91.52 ± 0.06	91.42 ± 0.10	91.29 ± 0.08	91.09 ± 0.09	90.82 ± 0.07
	Soft shadow	91.58 ± 0.07	91.48 ± 0.08	91.22 ± 0.11	90.91 ± 0.07	90.54 ± 0.10	90.10 ± 0.09
	Gaussian blur	91.58 ± 0.07	89.78 ± 0.10	79.94 ± 0.13	56.58 ± 0.11	39.11 ± 0.15	22.30 ± 0.12
	Misregistration	91.58 ± 0.07	90.80 ± 0.07	88.97 ± 0.09	84.25 ± 0.12	79.50 ± 0.08	74.80 ± 0.11
S2Looking	Brightness	65.78 ± 0.67	65.63 ± 0.59	65.38 ± 0.61	64.92 ± 0.58	64.31 ± 0.62	63.62 ± 0.60
	Hue	65.78 ± 0.67	65.70 ± 0.63	65.55 ± 0.57	65.32 ± 0.60	65.04 ± 0.59	64.72 ± 0.61
	Soft shadow	65.78 ± 0.67	65.66 ± 0.58	65.44 ± 0.62	65.08 ± 0.56	64.67 ± 0.61	64.18 ± 0.59
	Gaussian blur	65.78 ± 0.67	64.92 ± 0.62	62.41 ± 0.60	57.84 ± 0.63	51.06 ± 0.58	42.18 ± 0.62
	Misregistration	65.78 ± 0.67	65.24 ± 0.57	64.56 ± 0.56	63.42 ± 0.59	61.96 ± 0.63	60.24 ± 0.58
SYSU-CD	Brightness	84.46 ± 0.08	84.22 ± 0.10	83.91 ± 0.07	83.55 ± 0.11	82.96 ± 0.09	82.22 ± 0.12
	Hue	84.46 ± 0.08	84.32 ± 0.06	84.16 ± 0.10	83.96 ± 0.08	83.70 ± 0.12	83.38 ± 0.09
	Soft shadow	84.46 ± 0.08	84.25 ± 0.09	84.02 ± 0.11	83.73 ± 0.07	83.28 ± 0.10	82.71 ± 0.13
	Gaussian blur	84.46 ± 0.08	82.73 ± 0.12	79.06 ± 0.15	73.10 ± 0.13	65.48 ± 0.17	56.62 ± 0.14
	Misregistration	84.46 ± 0.08	84.07 ± 0.08	83.52 ± 0.12	82.74 ± 0.10	81.70 ± 0.14	80.24 ± 0.11

4.5. Ablation Studies

The preceding analyses reveal different behaviors in boundary localization, sparse-change detection, and appearance perturbation. The following ablations relate these observations to specific design choices by examining the front-end change representation, expert organization, individual experts, expert-pool capacity, the Adaptive Expert's dilation configuration, and the training constraints separately.

4.5.1. Bias Correction and Change Representation

Bias correction is designed to reduce cross-temporal statistical offsets while preserving local differences associated with genuine change. To distinguish this task-specific correction from generic feature normalization, Table 12 compares No Correction, Instance Normalization (IN), AdaIN, and the proposed Residual Correction under the same clean and perturbed conditions.

Table 12. Comparison with normalization-based alternatives. Perturbed columns report the controlled S5 setting. The best value in each metric is marked in bold.

Dataset	Method	Clean	Brightness S5	Soft Shadow S5
LEVIR-CD+	No Correction	90.88 ± 0.08	85.90	86.89
	Instance Normalization	90.51 ± 0.20	87.70	88.19
	AdaIN	90.66 ± 0.19	87.47	88.06
	Proposed Residual Correction	**91.58 ± 0.07**	**89.39**	**90.10**
SYSU-CD	No Correction	82.94 ± 0.14	78.03	78.81
	Instance Normalization	82.67 ± 0.26	79.96	80.34
	AdaIN	82.54 ± 0.27	79.52	79.98
	Proposed Residual Correction	**84.46 ± 0.08**	**82.22**	**82.71**

Under brightness and shadow perturbations, IN and AdaIN recover part of the performance loss relative to No Correction, but both yield lower clean-condition F1 than Residual Correction and remain below it under the perturbed conditions. Generic normalization alone therefore does not preserve the original change discrimination and correct cross-temporal appearance drift as effectively as the proposed residual formulation. Table 13 then separates change-representation construction from statistical correction: w/o Bias Correction retains the complete bitemporal change representation but removes the affine correction, whereas ΔF Only keeps only direct bitemporal feature differencing.

Table 13. Ablation results for Bias Correction and change representation (mean ± std, %). The best value in each metric is marked in bold.

	SYSU-CD		LEVIR-CD+	
Variant	**F1**	**IoU**	**F1**	**IoU**
Full SARE-CD	**84.46 ± 0.08**	**73.10 ± 0.22**	**91.58 ± 0.07**	**84.47 ± 0.19**
w/o Bias Correction	82.94 ± 0.14	70.86 ± 0.22	90.88 ± 0.08	83.28 ± 0.18
ΔF Only	82.66 ± 0.19	70.44 ± 0.22	90.62 ± 0.15	82.84 ± 0.11

The learned affine parameters also reflect the response of the correction module to increasing radiometric drift. As brightness severity increases from S1 to S3 and S5, the mean absolute responses of γ_l / β_l increase from 0.0041/0.0020 to 0.0122/0.0058 and 0.0209/0.0101. On LEVIR-CD+ unchanged pixels at S5, the normalized channel-mean drift decreases from 0.033181 before correction to 0.023473 after correction, while the channel-standard-deviation drift decreases from 0.029853 to 0.014189. On large-change subsets with $r \geq 20\%$, the full model improves F1 from 92.86% to 93.80% on LEVIR-CD+ and from 87.50% to 88.33% on SYSU-CD. The correction therefore becomes more responsive as radiometric drift increases while preserving discriminative information for large genuine changes in the tested cases.

4.5.2. Heterogeneous Expert Design

To isolate the contribution of functional heterogeneity, Table 14 keeps the front-end change representation and training constraints fixed while changing the expert organization. Homogeneous Dense-Gating MoE gives all three experts the same structure, whereas the Parameter-Matched Shared Path has comparable model capacity but removes expert specialization and processes the complete change representation through one path.

Table 14. Ablation results for heterogeneous expert design (mean ± std, %). The best value in each metric is marked in bold.

	SYSU-CD		LEVIR-CD+	
Variant	**F1**	**IoU**	**F1**	**IoU**
Heterogeneous Experts (SARE-CD)	**84.46 ± 0.08**	**73.10 ± 0.22**	**91.58 ± 0.07**	**84.47 ± 0.19**
Homogeneous Dense-Gating MoE	83.40 ± 0.12	71.52 ± 0.16	90.96 ± 0.13	83.42 ± 0.15
Parameter-Matched Shared Path	83.28 ± 0.13	71.35 ± 0.12	90.99 ± 0.17	83.47 ± 0.15

The homogeneous dense-gating variant retains multi-branch soft routing but yields lower F1 and IoU than SARE-CD. With 15.38 M parameters and 20.12 G FLOPs, the Parameter-Matched Shared Path achieves F1 scores of 83.28 ± 0.13, 90.99 ± 0.17, and 64.47 ± 0.36 on SYSU-CD, LEVIR-CD+, and S2Looking, respectively, remaining below the full model on all three datasets.

4.5.3. Contribution of Individual Experts

Table 15 presents the ablation results for removing individual experts. Removing any expert reduces F1 and IoU on both datasets, confirming that the three branches provide complementary contributions. The largest decrease on SYSU-CD follows removal of the Structure Expert, while LEVIR-CD+ shows consistent losses for all three removals.

Table 15. Ablation results for removing individual experts (mean ± std, %). The best value in each metric is marked in bold.

	SYSU-CD		LEVIR-CD+	
Variant	**F1**	**IoU**	**F1**	**IoU**
Full SARE-CD	**84.46 ± 0.08**	**73.10 ± 0.22**	**91.58 ± 0.07**	**84.47 ± 0.19**
w/o Structure Expert	82.99 ± 0.12	70.93 ± 0.26	90.86 ± 0.08	83.26 ± 0.18
w/o Appearance Expert	83.06 ± 0.12	71.03 ± 0.30	90.84 ± 0.07	83.21 ± 0.18
w/o Adaptive Expert	83.42 ± 0.19	71.56 ± 0.16	90.88 ± 0.07	83.28 ± 0.11

4.5.4. Expert-Pool Capacity

Table 16 reports the expert-pool capacity results. Increasing the expert count beyond the three functionally defined branches does not improve performance. The K = 4 and K = 5 variants are lower on both datasets, indicating that adding capacity without a new functional prior is not beneficial under the current routing design.

Table 16. Ablation results for expert-pool capacity (mean ± std, %). The best value in each metric is marked in bold.

	SYSU-CD		LEVIR-CD+	
Variant	**F1**	**IoU**	**F1**	**IoU**
Adaptive + Structure + Appearance (K = 3)	**84.46 ± 0.08**	**73.10 ± 0.22**	**91.58 ± 0.07**	**84.47 ± 0.19**
K = 3 + 1 Additional Expert (K = 4)	83.11 ± 0.16	71.11 ± 0.26	90.92 ± 0.07	83.35 ± 0.15
K = 3 + 2 Additional Experts (K = 5)	83.12 ± 0.12	71.12 ± 0.22	90.77 ± 0.08	83.10 ± 0.18

4.5.5. Dilated-Convolution Configuration

Table 17 compares different dilated-convolution configurations. The dilation pair (2,3) provides the strongest results on both datasets. The remaining settings are lower by different margins, showing that the Adaptive Expert benefits from combining two receptive-field scales rather than simply enlarging the dilation rate.

Table 17. Ablation results for different dilated-convolution configurations (mean ± std, %). The best value in each metric is marked in bold.

	SYSU-CD		LEVIR-CD+	
Variant	**F1**	**IoU**	**F1**	**IoU**
(2,3) (Ours)	**84.46 ± 0.08**	**73.10 ± 0.22**	**91.58 ± 0.07**	**84.47 ± 0.19**
(2,2)	83.29 ± 0.22	71.37 ± 0.19	90.88 ± 0.08	83.28 ± 0.09
(3,3)	82.79 ± 0.19	70.63 ± 0.22	91.04 ± 0.07	83.56 ± 0.11
(1,2)	83.09 ± 0.10	71.08 ± 0.16	90.85 ± 0.13	83.23 ± 0.11
(2,4)	82.55 ± 0.10	70.29 ± 0.30	90.89 ± 0.15	83.30 ± 0.15
(1,1) Standard Convolution	83.35 ± 0.14	71.45 ± 0.26	90.96 ± 0.13	83.42 ± 0.12
Single-layer, d = 2	82.94 ± 0.14	70.85 ± 0.22	90.75 ± 0.07	83.06 ± 0.12
Single-layer, d = 3	83.13 ± 0.10	71.13 ± 0.26	90.92 ± 0.13	83.35 ± 0.09

4.5.6. Training-Loss Constraints

Table 18 shows that deep supervision, routing entropy, and load balancing all contribute to the final optimization. The learned routing distributions further distinguish the roles of the two routing constraints. On LEVIR-CD+, the full model has a mean routing entropy of 0.183 and a load variance of 4.97×10^{-4}; removing $\mathcal{L}_{\text{ent}}$ increases the entropy to 0.847, whereas removing $\mathcal{L}_{bal}$ increases the load variance to 3.9413×10^{-2}. On SYSU-CD, the corresponding full-model values are 0.201 and 8.06×10^{-4}, increasing to 0.902 without $\mathcal{L}_{\text{ent}}$ and 2.5022×10^{-2} without $\mathcal{L}_{bal}$. These changes are consistent with the intended roles

of the two terms: $\mathcal{L}_{\text{ent}}$ primarily controls the concentration of pixel-wise assignments, whereas $\mathcal{L}_{bal}$ limits imbalance in overall expert usage.

Table 18. Ablation results for training-loss constraints (mean ± std, %). The best value in each metric is marked in bold.

	SYSU-CD		LEVIR-CD+	
Variant	**F1**	**IoU**	**F1**	**IoU**
Full Loss	**84.46 ± 0.08**	**73.10 ± 0.22**	**91.58 ± 0.07**	**84.47 ± 0.19**
w/o Deep Supervision	83.02 ± 0.12	70.97 ± 0.16	90.74 ± 0.07	83.05 ± 0.20
w/o Load Balancing	82.87 ± 0.19	70.75 ± 0.14	90.83 ± 0.13	83.20 ± 0.12
w/o Routing Entropy	83.24 ± 0.19	71.29 ± 0.14	90.98 ± 0.13	83.46 ± 0.20
Change Loss Only	83.06 ± 0.22	71.03 ± 0.26	90.96 ± 0.09	83.42 ± 0.15

Using the same default routing configuration, we also extend the key-module removals to S2Looking. The full model reaches 65.78 ± 0.67 F1; removing Bias Correction, the Structure Expert, the Appearance Expert, the Adaptive Expert, deep supervision, load balancing, and routing entropy yields 64.86 ± 0.31, 64.73 ± 0.34, 64.35 ± 0.33, 64.92 ± 0.30, 64.68 ± 0.35, 64.51 ± 0.36, and 64.89 ± 0.32, respectively. The consistent direction of these changes across LEVIR-CD+, S2Looking, and SYSU-CD indicates that the contributions of the key components are not confined to a single dataset.

4.5.7. Routing-Temperature Sensitivity

The routing temperature controls the sharpness of the pixel-wise expert distribution in Equation (12). We vary $\tau \in \{0.25, 0.50, 1.00, 2.00, 4.00\}$ while keeping the remaining network architecture and training settings unchanged. The resulting F1 scores on SYSU-CD and LEVIR-CD+ are reported in Table 19.

Table 19. Sensitivity to routing temperature τ (F1 as mean ± std over three independent runs, %). The best value in each metric is marked in bold.

τ	**SYSU-CD F1**	**LEVIR-CD+ F1**
0.25	83.31 ± 0.13	90.86 ± 0.11
0.50	83.96 ± 0.10	91.20 ± 0.09
1.00 (Default)	**84.46 ± 0.08**	**91.58 ± 0.07**
2.00	83.88 ± 0.11	91.13 ± 0.10
4.00	83.12 ± 0.14	90.71 ± 0.12

The default setting $\tau = 1.0$ yields the highest mean F1 on both datasets. Within the moderate range $\tau \in [0.5, 2.0]$, the largest decreases relative to $\tau = 1.0$ are 0.58 percentage points on SYSU-CD and 0.45 percentage points on LEVIR-CD+, whereas the more extreme settings $\tau = 0.25$ and 4.0 produce larger degradation.

5. Discussion

5.1. Routing-Weight Visualization and Expert Specialization

Figure 8 visualizes the continuous routing weights of the full model across different scenes. The three experts form distinct spatial response patterns around building boundaries, shadow- and texture-affected regions, and locations where structural and appearance cues overlap. Because routing is dense, these maps represent the relative fusion preferences among three already-computed expert features rather than the activation or deactivation of expert branches.

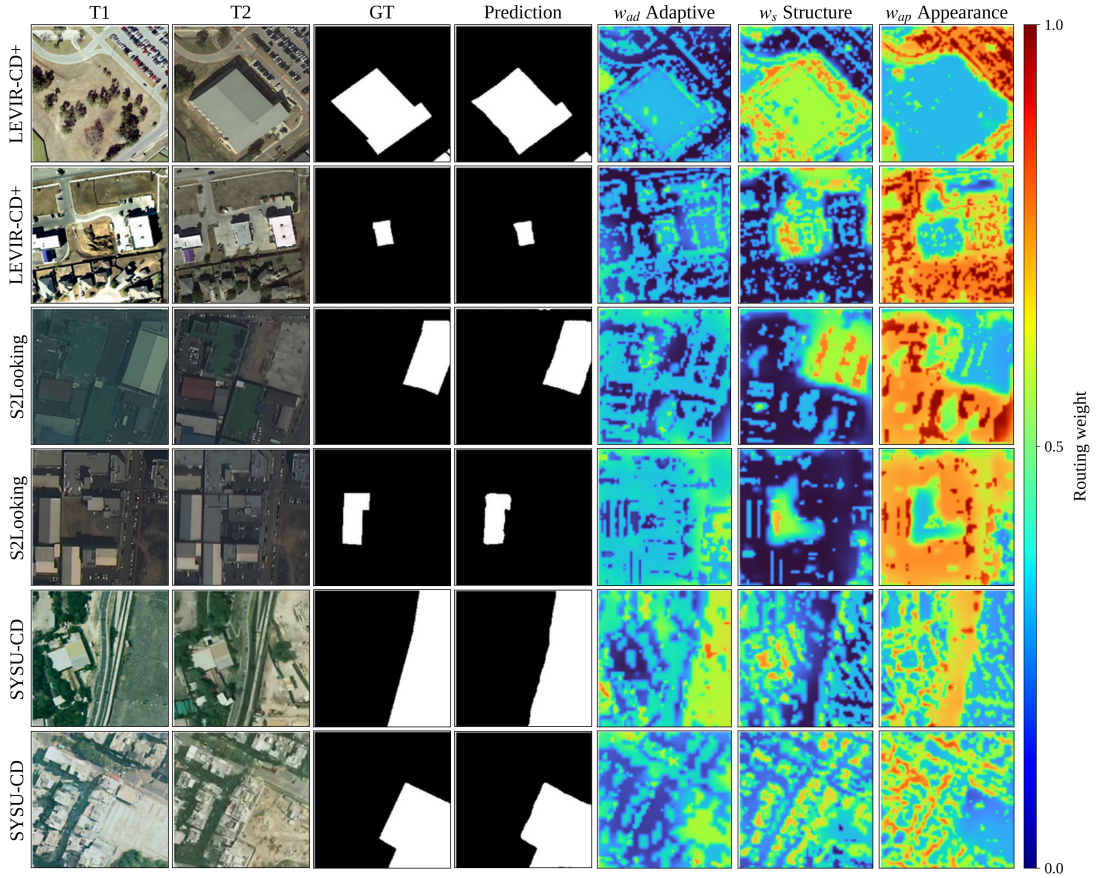

Figure 8. Routing-weight visualization across datasets. Each pair of rows corresponds to LEVIR-CD+, S2Looking, and SYSU-CD, respectively. From left to right, the columns show the bitemporal images I_1 and I_2, the ground truth, the prediction, and the routing weights of the Adaptive, Structure, and Appearance Experts. All routing-weight maps share the same color scale from 0 to 1.

Figure 9 further compares the dominant-expert distributions of the full model, the model without routing entropy, and the model without load balancing. Removing $\mathcal{L}_{\text{ent}}$ produces more fragmented dominant regions and weaker local concentration, whereas removing $\mathcal{L}_{bal}$ shifts the assignments toward fewer experts. These spatial changes are consistent with the routing-entropy and load-variance statistics in Section 4.5.6, indicating that the two constraints mainly affect local routing concentration and overall expert-usage balance, respectively.

We do not enforce a rigid assignment where a specific expert must solely handle a predetermined type of difference. Instead, under the data-driven dense soft-gating mechanism, the different experts exhibit significant region-conditioned routing preferences.

To provide direct statistical evidence, we extracted the continuous expert weights across three mutually exclusive, pre-defined regions: the ground-truth change boundary zone, high-frequency appearance difference regions, and stable unchanged regions. A bootstrap resampling analysis (10,000 iterations) with image-level pairing revealed the following: (1) Within the ground-truth boundary zones, the average weight assigned to the Structure Expert is significantly higher than its weight in stable regions (e.g., on LEVIR-CD+, the mean difference is $+0.188$, with a 95% confidence interval (CI) of $[0.154, 0.222]$). (2) Within appearance difference regions, the Appearance Expert's average weight significantly increases (on LEVIR-CD+, the mean difference is $+0.149$, with a 95% CI of $[0.121, 0.177]$). Because none of these confidence intervals cross zero, there is solid statistical support that the different experts, guided by their complementary architectural priors, form

differentiated routing collaborations in regions dominated by their corresponding features. Table 20 summarizes the continuous expert weights within these predefined regions.

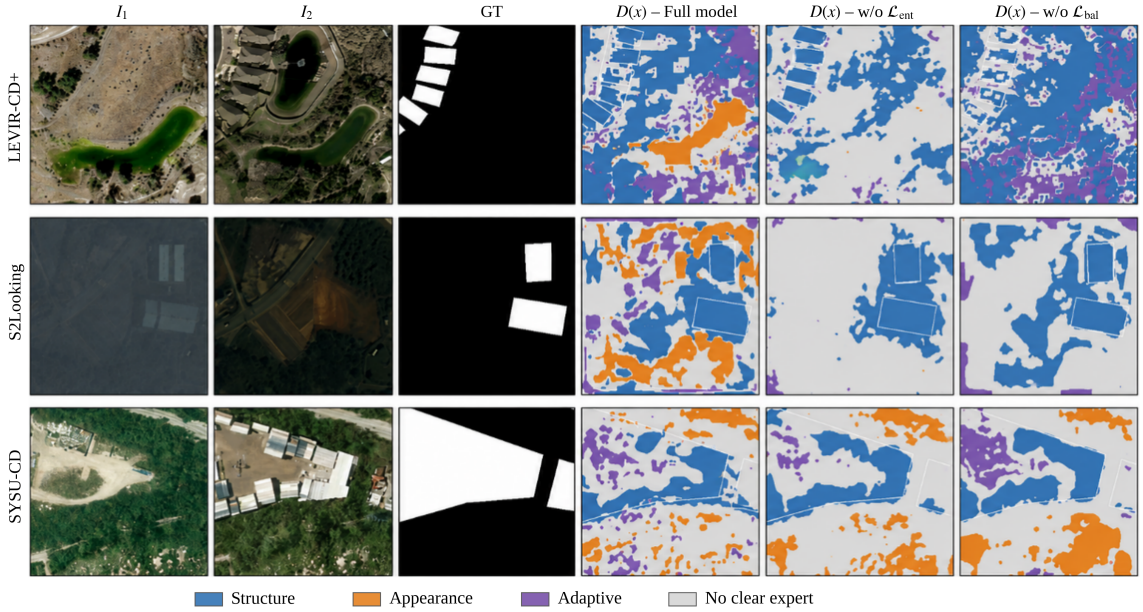

Figure 9. Dominant-expert distributions under routing-constraint ablations. Each row shows, from left to right, the bitemporal images, the ground truth, the dominant-expert maps of the full model, the variant without routing entropy (w/o $\mathcal{L}_{\text{ent}}$), and the variant without load balancing (w/o $\mathcal{L}_{bal}$). Purple, blue and orange indicate Adaptive, Structure and Appearance dominance, respectively, and gray indicates that no expert is clearly dominant.

Table 20. Continuous expert weights within pre-defined regions (image-level mean ± std).

Dataset	Region	Adaptive	Structure	Appearance
LEVIR-CD+	GT Boundary Zone	0.238 ± 0.071	0.526 ± 0.092	0.236 ± 0.066
	Appearance Diff.	0.270 ± 0.063	0.248 ± 0.057	0.482 ± 0.084
	Stable Unchanged	0.329 ± 0.041	0.338 ± 0.044	0.333 ± 0.042
S2Looking	GT Boundary Zone	0.274 ± 0.083	0.471 ± 0.101	0.255 ± 0.079
	Appearance Diff.	0.294 ± 0.076	0.239 ± 0.061	0.467 ± 0.093
	Stable Unchanged	0.334 ± 0.049	0.335 ± 0.052	0.331 ± 0.050
SYSU-CD	GT Boundary Zone	0.282 ± 0.088	0.452 ± 0.107	0.266 ± 0.082
	Appearance Diff.	0.316 ± 0.081	0.254 ± 0.067	0.430 ± 0.095
	Stable Unchanged	0.337 ± 0.054	0.334 ± 0.057	0.329 ± 0.053

5.2. Sparse-Change Failure Cases and Future Directions

The pooled statistics in Table 7 identify the SYSU-CD Small subset as a difficult operating regime: SARE-CD reaches 64.16% Recall and 35.21% Precision, corresponding to 45.47% F1 and 29.42% IoU. To examine the routing pattern associated with these errors, the Failure and Successful groups are formed from the same SYSU-CD Small candidate pool using fixed sample-level criteria before any routing statistics are computed. A sample is classified as a Failure case when its false-positive pixel count exceeds its correctly detected change pixel count ($FP > TP$), and as a Successful case when its correctly detected change pixel count exceeds its false-positive pixel count ($TP \geq FP$). No sample is selected manually. The same grouping is then used to compute routing weights over ground-truth changed, false-positive, and true-negative regions.

On ground-truth changed pixels, the Successful group assigns a higher mean weight to the Structure Expert than the Failure group (0.540 versus 0.420). In false-positive regions of the Failure group, the Structure weight decreases to 0.267 while the Adaptive weight

increases to 0.393; the Appearance weight is 0.340, close to 0.346 in the Successful group. The dominant difference is therefore associated with weaker structural routing together with stronger Adaptive response, rather than with an isolated increase of the Appearance branch. These measurements describe a statistical association between sparse-scene errors and routing behavior, not a causal attribution of false positives to a single expert. Table 21 summarizes the routing-weight statistics for the two groups across the three pixel regions.

Table 21. Routing-weight statistics for sparse-change samples on SYSU-CD (mean ± pixel-level std).

Sample Group	Pixel Region	Adaptive	Structure	Appearance
Failure cases	GT-change pixels	0.280 ± 0.41	0.420 ± 0.45	0.300 ± 0.42
	False-positive pixels	0.393 ± 0.45	0.267 ± 0.40	0.340 ± 0.43
	True-negative pixels	0.320 ± 0.42	0.364 ± 0.44	0.316 ± 0.42
Successful cases	GT-change pixels	0.220 ± 0.37	0.540 ± 0.45	0.240 ± 0.38
	False-positive pixels	0.356 ± 0.44	0.298 ± 0.41	0.346 ± 0.43
	True-negative pixels	0.312 ± 0.36	0.380 ± 0.42	0.308 ± 0.39

These results indicate that sparse targets in cluttered backgrounds still require more stable structural evidence and stronger suppression of non-target responses. The perturbation experiments further show that strong Gaussian blur and larger spatial displacement directly damage local structural information and remain difficult conditions for the current model. The present evaluation is limited to bitemporal optical building change detection; multitemporal sequences, multisensor data, and cross-region transfer have not yet been examined.

6. Conclusions

SARE-CD achieves F1 scores of 91.58%, 65.78%, and 84.46% on LEVIR-CD+, S2Looking, and SYSU-CD, respectively. In the boundary evaluation, its BF1 and BIoU are higher than those of the two comparison methods on all three datasets. The appearance-oriented ablations further show that the Appearance Expert with channel attention reduces the mean change confidence on unchanged pixels while maintaining or improving F1 through a better precision–recall balance. Controlled perturbation tests show stable performance over the tested ranges of brightness, hue, and soft-shadow variation, whereas strong Gaussian blur and larger spatial misregistration cause substantially larger degradation. The ablation studies further verify the contributions of bias correction, heterogeneous expert specialization, expert-pool design, the Adaptive Expert's dilation configuration, and the routing constraints.

The experiments also define the current scope of the evidence. The relatively low precision on the SYSU-CD sparse-change subset indicates that false positives in complex backgrounds remain a major error source, while strong blur and larger spatial displacement weaken the spatial structure required for reliable change discrimination. The present conclusions are therefore restricted to the three bitemporal optical building-change datasets and the tested perturbation ranges. Future work will focus on suppressing sparse-target false positives, extending the framework to multitemporal and multisensor settings, and improving cross-region generalization.

Specifically, on the SYSU-CD sparse-change subset, SARE-CD's F1 is approximately 5.2–5.3 percentage points lower than ChangeMamba and SA-CDNet, mainly due to false positives in complex backgrounds.

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Abbreviations

The following abbreviations are used in this manuscript:

SARE-CD	Structure–Appearance Routed Experts for Building Change Detection
CD	Change Detection
BCE	Binary Cross-Entropy
MoE	Mixture of Experts
MLP	Multilayer Perceptron
IN	Instance Normalization
AdaIN	Adaptive Instance Normalization
CA	Channel Attention
OA	Overall Accuracy
F1	F1 Score
IoU	Intersection over Union
BF1	Boundary F1 Score
BIoU	Boundary Intersection over Union
GT	Ground Truth
TP	True Positive
FP	False Positive
FN	False Negative
TN	True Negative
ReLU	Rectified Linear Unit
GAP	Global Average Pooling
FLOPs	Floating-Point Operations
FMA	Fused Multiply-Add
FPS	Frames per Second
HSV	Hue–Saturation–Value
FP32	32-bit Floating Point
GPU	Graphics Processing Unit
CUDA	Compute Unified Device Architecture
CI	Confidence Interval

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